

Linear Regression & Machine Learning Master Guide (Telugu Transliteration)

Ee document lo, "Linear_regressionAlgorithmsNotes.pdf" lo unna concepts ni base chesukuni, meeku Machine Learning (ML) meeda **complete command** ravadaniki oka master guide prepare chesanu. Prathi concept ni pin-to-pin Telugu transliteration lo (English letters tho), nija jeevitham lo udaharanaalato, math formulas tho, mariyu **nidra lo lechina code rayagalige antha strong coding practice** tho explain chestanu.

1. Machine Learning Parichayam (What is Machine Learning?)

Machine Learning ante enti?

Machine Learning ante computer ki prathi rule ni explicit ga (manam cheppi) program cheyakunda, **data ni chusi dane antantata tana anubhavam tho nerchune vidhanam**.

- **Traditional Programming:** Rules + Data $\rightarrow$ Output
- **Machine Learning:** Data + Output $\rightarrow$ Rules (Computer nerchukuntundi!)

Nija Jeevitham lo Upayogalu (Real-Life Applications)

- **Personalized Feed Ranking:** Instagram leda YouTube lo meeku nachhe videos ni chupinchatam.
- **Video Recommendation System:** Netflix lo next em chudali ani suggest cheyatam.
- **Auto Content Moderation:** Bad comments leda spam ni automatically delete cheyatam.
- **Face Filters:** Snapchat leda Instagram filters lo face ni track cheyatam.
- **Fake Account Detection:** Bank transactions lo leda social media lo fraud accounts ni gurthupattatam.

Machine Learning Vidhalu (Types of Machine Learning)

ML Type	Artham (Meaning)	Nija Jeevitham lo Udaharana (Real-Life Example)
1. Supervised Learning	Labelled data (Input mariyu correct output rendu untayi) tho model train avtundi.	Study hours (input) → Marks (output). House size → Price.
2. Unsupervised Learning	Labels leni raw data istaru. Model dane patterns ni gurthupattukuntundi.	Customer buying behavior batti groups ga divide cheyatam (Customer Segmentation).
3. Semi-Supervised Learning	Konchem labelled data mariyu chala unlabelled data untayi.	Medical imaging lo konni images ki disease label untundi, chala images ki undavu.
4. Reinforcement Learning	Reward (bhumarsham) mariyu Punishment (shiksha) dwara computer trial and error tho nerchukuntundi.	Self-driving cars road meeda nadavadam, Chess game adede AI.

2. Supervised Learning: Regression vs Classification

Supervised learning lo rendu mukhya divisions unnayi:

- 1. **Regression:** Output oka **number (continuous value)** ayithe, dani Regression antaru. (Udaharana: Illu entha viluva chestundi? Salary entha? Temperature entha?).
- 2. **Classification:** Output oka **category leda label** ayithe, dani Classification antaru. (Udaharana: Email spam aa? Kada? Disease unda? Leda? Pass aa? Fail aa?).

3. Linear Regression Deep-Dive (Linear Regression ante Enti?)

Linear Regression ante oka input (X) mariyu output (Y) madhya unna straight-line relationship ni use chesi, oka kotha number ni predict chese method.

Input Feature X: House Size

Linear Regression Model

Predicted Output Y: House Price

Linear Regression Equation

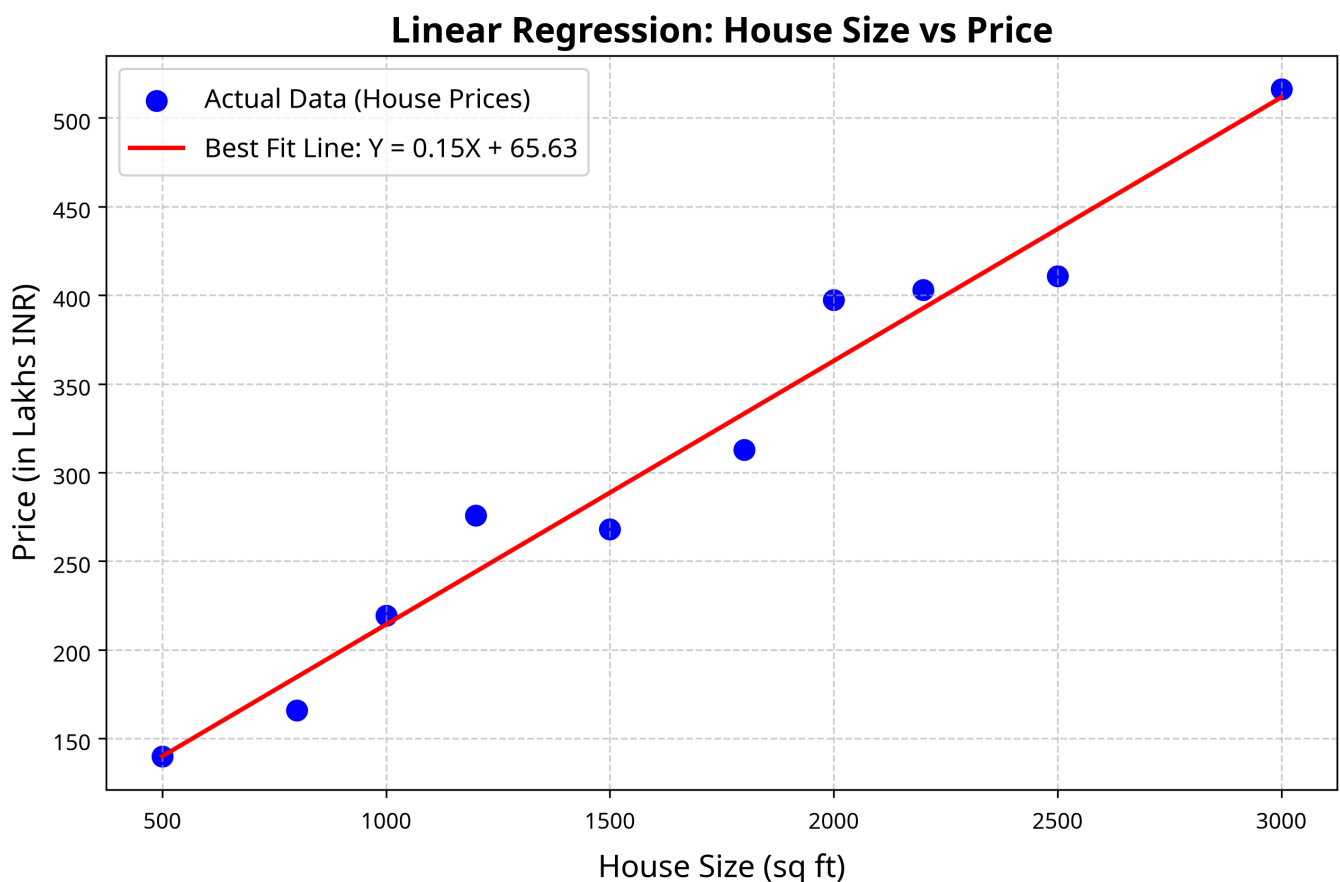
Model oka straight line ni draw chestundi:

$$Y = mX + b$$

- **Y (Dependent Variable):** Manam predict cheyalanukunna output (Udaharana: Price).
- **X (Independent Variable):** Manam input ga iche feature (Udaharana: House size).
- **m (Slope / Weight):** X oka unit perigithe, Y entha marutundi ani cheptundi.
- **b (Intercept / Bias):** X = 0 unnapudu Y ki unna starting value.

Vizualization: House Size vs House Price

Nija prapancham lo data points ela untayo, mariyu linear regression line vatini ela connect chestundo kinda chudandi:



Linear Regression Conditions (Eppudu Vadali?)

- Data lo relationship oka **straight-line pattern** lo undali.
- X perigithe Y steady ga peragali leda taggali.
- Outliers (chala dooramga unna values) undakudadu.
- Zero Multicollinearity (features madhya extra correlation undakudadu).

4. Regression Loss Functions (Thappu ni Ela Kolavali?)

Model predict chesina value ki, actual value ki madhya unna difference ne **Loss (Error)** antaru. Ee error ni measure cheyadaniki rendu mukhya methods unnayi:

1. Mean Squared Error (MSE)

Prathi error ni square chesi, tarwata average teesukuntaru.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- Mukhyamaina vishayam:** Squares cheyadam valla, **pedda errors ki chala heavy penalty** padutundi. Anduke idi **Outliers** ki chala sensitive ga untundi.

2. Mean Absolute Error (MAE)

Errors yokka absolute values ni teesukuni average chestaru (squares cheyaru).

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- Mukhyamaina vishayam:** Idi squares cheyadu kabatti, **outliers unna kooda pedda effect chupadhu.**

Loss Function	Sensitivity to Outliers	Formula Feature
MSE	Chala sensitive (Pedda errors ki heavy penalty)	Squared differences
MAE	Less sensitive (Normal errors laage treat chestundi)	Absolute differences

5. Model Evaluation Metrics (Model Entha Bagundi?)

Model train ayina tarwata adi entha бага pani chestundi ani kolavadaniki evaluation metrics use chestaru.

R^2 Score (Coefficient of Determination)

- Mee model data ni entha бага fit chesindo cheptundi.
- Value range: **0 to 1** (leda 0% nunchi 100%).
- **1** ante perfect prediction. **0** ante model em nerchukoledu (random guess).
- *Udaharana:* $R^2 = 0.80$ ayithe, mee model data lo unna variation lo **80% ni successfully explain chestundi** ani ardam!

Adjusted R^2

- R^2 ki oka bad habit undi: manam enni useless features add chesina R^2 perugutundi leda alage untundi.
- **Adjusted R^2** daanini correct chestundi. Idi useless features add chesthe **penalty** istundi, anduke Adjusted R^2 eppudu $\leq R^2$ untundi.

6. MASTER CODING SECTION: Step-by-Step Python Implementation

Meeru nidra lo lechina error lekunda code rayadaniki, ikkada **10-Step Complete Python Code** ichchanu. Prathi step enduku rastunnamo clear ga comment chesanu.

```

# =====
# STEP 1: Import Required Libraries
# =====
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

# =====
# STEP 2 & 3: Load Dataset & Preprocessing
# =====
#manam ikkada oka sample housing dataset ni create chestunnamu (Real project)
np.random.seed(42)
house_sizes = np.random.randint(500, 3500, 100) # Input X (Square feet)
# Price = 50 + 0.12 * size + random noise
house_prices = 50 + 0.12 * house_sizes + np.random.normal(0, 20, 100) # Output Y

df = pd.DataFrame({'Size': house_sizes, 'Price': house_prices})

# Missing values check and cleaning (Preprocessing)
print("Missing values in dataset:", df.isnull().sum().sum())

# =====
# STEP 4 & 5: EDA & Feature Engineering
# =====
# Data ni observe cheyadam mariyu features ni prepare cheyadam
print(df.head())

# =====
# STEP 6: Divide Data into X (Input) and Y (Target)
# =====
X = df[['Size']] # Input feature (2D array kavali scikit-learn ki)
Y = df['Price'] # Target variable (1D array)

# =====
# STEP 7: Train-Test Split
# =====
# Moktham data ni model ki isthe, adi memorize chestundi. Anduke train ma
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=42)

print(f"Training data size: {X_train.shape[0]} rows")

```

```

print(f"Testing data size: {X_test.shape[0]} rows")

# =====
# STEP 8: Train Model (Linear Regression)
# =====
# 1. Model ni import chesam (paina chesam)
# 2. Model object ni create chestam
model = LinearRegression()

# 3. Model ni training data tho fit chestam (Model nerchukuntundi: m mari
model.fit(X_train, Y_train)

# Model nerchukunna parameters ni chudandi:
print("Learned Slope (m):", model.coef_[0])
print("Learned Intercept (b):", model.intercept_)

# 4. X_test meeda predictions chestam
Y_pred = model.predict(X_test)

# =====
# STEP 9: Evaluate Model (Loss & Metrics)
# =====
mse = mean_squared_error(Y_test, Y_pred)
mae = mean_absolute_error(Y_test, Y_pred)
r2 = r2_score(Y_test, Y_pred)

print("\n--- MODEL EVALUATION RESULTS ---")
print(f"Mean Squared Error (MSE): {mse:.2f}")
print(f"Mean Absolute Error (MAE): {mae:.2f}")
print(f"R2 Score: {r2:.4f}")

# =====
# STEP 10: Saving Model (Optional for Production)
# =====
import pickle
with open('linear_regression_model.pkl', 'wb') as file:
    pickle.dump(model, file)
print("\nModel successfully saved to 'linear_regression_model.pkl'!")

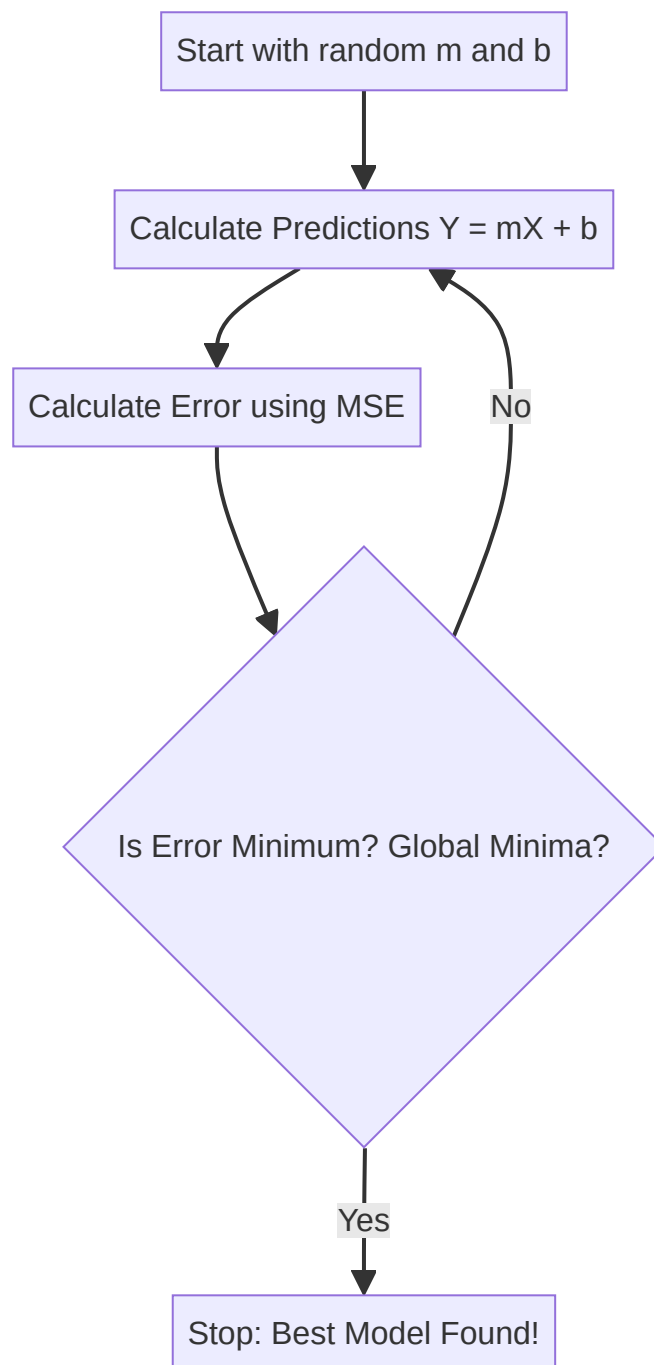
```

7. Gradient Descent Technique (Error ni Ela Tagginchali?)

Model ki m mariyu b values modalata teliyavu. Adi random ga start chesi, error ni step-by-step tagginchukuntu potundi. Ee optimization process ne **Gradient Descent** antaru.

Simple Analogy: Fog lo Konda Digadam

Meeru oka konda meeda kattukuni unnaru, valley (lothu pradesam) ki cherukovali. Meeru kallu kattukuni unnaru kabatti, kaalla kinda slope ela undi ani feel ayyi, eppudu **steep ga dige direction** lo chinna chinna steps vestaru. Ide Gradient Descent!



Learning Rate (Step Size / α)

- **Too Big (Chala pedda step):** Minimum ni datukuni avathaliki vellipothundi (Overshooting, unstable).
 - **Too Small (Chala chinna step):** Minimum cherukodaaniki chala rojulu pattochu (Very slow learning).
 - **Most Common Value:** 0.01 (Chala manchi starting point).
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References:

1. [Linear_regressionAlgorithmsNotes.pdf](#) - Original PDF document for Machine Learning and Linear Regression notes.